Call Me Maybe? Exposing Patterns of Shadow Scam Ecosystems via Open-Source Victim Complaints

ABSTRACT

Existing defenses against telephony fraud operate on individual phone numbers. Blacklists rank them by complaint volume, caller-ID authentication verifies the displayed number, and carrier abuse teams investigate reported numbers one at a time. But scam operations are multi-number organizations that provision disposable frontline numbers separately from persistent backend callback hubs. A defense that sees only the frontline misses the backend entirely. However, victim complaint narratives often contain more than a single phone number, including callback numbers, spoofed caller-ID values, and other numbers encountered during the interaction. These cross-references have been largely overlooked. To this end, we introduce **EXPOSE**, a framework that uncovers hidden scam infrastructure from victim complaint narratives. Specifically, our framework ***(1)*** extracts phone-number cross-references from complaint text; ***(2)*** connects frontline numbers to hidden backend targets through a directed graph; ***(3)*** enriches callback targets with carrier metadata; and ultimately ***(4)*** exposes scamming campaigns and shadow scam ecosystems. We validate **EXPOSE** on a public 800notes corpus of 93,008 phone numbers (i.e., complaint page) spanning 19 years. The framework uncovers 666 callback scam numbers, of which 541 (81.2%) are lurking with zero direct complaints. Notably, 76.8% of the callback numbers are associated with toll-free and burner numbers, making them inherently difficult to trace. Moreover, **EXPOSE** exposes 203 persona-based scamming campaigns and 12 shadow scam ecosystems. Cross-validating the 666 callback scam numbers against 8.54 million Federal Trade Commission (FTC) complaints confirms that 59.2% of shadow callback targets are invisible to the federal complaint system as well. To support ongoing efforts in exposing and combating scams, we make the **EXPOSE** framework and its associated dataset publicly available.

CCS CONCEPTS

• **Security and privacy** → **Web security**;

KEYWORDS

Telephony fraud, Scam detection, Shadow infrastructure, Open-source intelligence

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1 INTRODUCTION

In 2024, U.S. consumers reported losing over $12.5 billion to fraud, a 25% increase over the prior year, with phone calls remaining the second most common contact method used by scammers [4]. Losses from imposter scams alone, which frequently rely on telephony as the primary delivery channel, totaled $2.95 billion, and consumers aged 60 and over reported $2.4 billion in losses, often initiated by phone [5]. These losses almost certainly underestimate the true scale. The Federal Trade Commission (FTC) estimates that actual losses experienced by older adults alone may be as high as $82 billion when accounting for underreporting [5].

Existing defenses treat each reported scam phone number independently. Blacklists flag numbers that accumulate complaints [15]. Secure Telephone Identity Revisited/Signature-based Handling of Asserted information using toKENs (STIR/SHAKEN), the federal caller-ID authentication framework, authenticates the caller ID attached to each call [3]. Carrier abuse teams investigate numbers reported to their own networks [23]. These defenses share a structural assumption that the phone number visible to the victim is the meaningful unit of analysis. But scam operations are not single numbers. They are multi-number organizations that provision disposable frontline numbers for victims to see, while keeping persistent backend callback hubs in the shadow. These backend numbers are what we call `shadow numbers` that no victim ever reports directly, that have no complaint page anywhere, but that victims describe in their complaint text when they write "they told me to call (877)-XXX-XXXX." A defense that sees only the frontline misses these shadow numbers entirely.

We argue that victim complaint narratives contain a signal that has been systematically overlooked. When victims report a phone number, they frequently mention other numbers in their complaint text, numbers they were told to call back, numbers that appeared on their caller ID, and numbers that texted them. These cross-references implicitly link the visible frontline to the hidden backend. No existing complaint system treats them as structured data.

We introduce **EXPOSE** (**Ex**posing **P**atterns **o**f **S**hadow Scam **E**cosystems), a framework that extracts these cross-references from complaint text and uses them to uncover hidden scam infrastructure. **EXPOSE** works in four stages. First, it scans complaint narratives for mentions of phone numbers and labels each by its role in the scam interaction (callback, spoofed caller ID, SMS, or other, Table 1). Second, it connects reported numbers to the mentioned numbers via a directed graph, revealing a large set of shadow phone numbers

that no victim ever reported directly. Third, it enriches the hidden layer with carrier and line-type metadata to characterize how the backend is provisioned. Fourth, it groups phone numbers into campaigns based on complaint-text similarity and links campaigns into shadow scam ecosystems through shared backend infrastructure.

This approach is entirely passive. It requires no interaction with scam operators, no carrier cooperation, and no honeypot infrastructure. Our framework is generalizable and operates entirely on publicly posted complaint data. Its input contract is minimal: a reported phone number, text narratives, and timestamps. We validate **EXPOSE** on a public 800notes[1] corpus of 93,008 reported phone numbers and 97,093 comments posted across their complaint pages, spanning 19 years. The framework uncovers 666 callback numbers, of which 541 (81.2%) are lurking, or shadow numbers never reported before. Toll-free numbers are 16.5× more common in the callback layer than in the reported corpus, and 85.3% of shadow toll-free callback numbers return a null carrier when queried through Twilio Lookup v2, a widely used carrier-identification Application Programming Interface (API)., making them opaque to standard traceback. Beyond retrospective analysis, **EXPOSE** can also flag hidden callback hubs as they emerge. When three unrelated victims, complaining about three different frontline numbers, all independently write "call back at" the same number, the framework treats that convergence as an alert. In our evaluation, every alert produced this way is a confirmed scam callback hub (100% precision), with a median delay of 6 days from the first mention to detection. Combining text similarity with shared backend evidence (common callback numbers and carrier metadata), we identify 203 persona-based campaigns and 12 shadow scam ecosystems.

We investigate three research questions:

- **RQ1** (Section 4): What hidden scam infrastructure can be uncovered from open-source victim complaints, and can cross-references between reported and mentioned phone numbers expose the shadow numbers behind them?
- **RQ2** (Section 5): How is the shadow callback layer provisioned, and can existing carrier-level defenses identify shadow scam infrastructure?
- **RQ3** (Section 6): Can correlating complaint text, callback relationships, and carrier metadata uncover higher-level shadow scam ecosystems?

This paper makes the following contributions:

- We show that open-source victim complaints contain cross-references exposing a large hidden layer of backend phone numbers invisible to every existing defense. Of 9,756 cross-reference observations, 81.1% point to shadow numbers with zero direct complaints, a publicly available signal overlooked as infrastructure intelligence.
- We design **EXPOSE**, a four-stage pipeline that extracts cross-reference relationships from raw complaint text and uses them to uncover hidden scam infrastructure. We validate it on a public 800notes corpus spanning 19 years and 93,008 reported phone numbers, and release the implementation as open-source `EXPOSE-Toolkit` to support reproducibility.
- We show the hidden callback layer is provisioned differently from the visible frontline. Toll-free numbers are 16.5× more common among callback numbers than reported ones, and 85.3% of shadow toll-free callbacks return no carrier identity under Twilio Lookup v2.
- We combine complaint-text similarity with shared callback and carrier evidence to recover 203 persona-based campaigns and 12 shadow scam ecosystems. Ten ecosystems are built solely from complaint-derived relationships, and 96.3% of their nodes fall outside every defense class we model.

The rest of the paper is organized as follows. Section 2 provides a telephony background and positions our work relative to prior studies. Section 3 describes the validation corpus and the four-stage pipeline. Sections 4 uncovers hidden scam infrastructure, Section 5 presents the carrier-level analysis of callback infrastructure, and Section 6 exposes the shadow scam ecosystem via a three-level hierarchy. Section 7 discusses limitations. Section 8 concludes our paper.

2 BACKGROUND AND RELATED WORK

This section provides the telephony-system background needed to understand the framework and positions our work relative to prior studies of telephony-based scams.

2.1 Phone Number Types and Carrier Identification

Telephone numbers in the United States, Canada, and several Caribbean jurisdictions follow the North American Numbering Plan (NANP) format NPA-NXX-XXXX, where the NPA is the area code, and the NXX identifies a central-office prefix [25]. Prefix assignments are recorded in the Local Exchange Routing Guide (LERG), which maps number blocks to assigned carriers. Because local number portability allows subscribers to move numbers across carriers, the current serving carrier can differ from the original assignee [23].

Three number types are relevant to this paper. Toll-free numbers use the 800, 888, 877, 866, 855, 844, and 833 area codes. They are administered through the SMS/800 system and managed by Responsible Organizations (RespOrgs) [3]. Toll-free numbers are legitimate business infrastructure, but they are also useful as geographically neutral callback endpoints because they carry no regional identity and are free for the caller to dial.

Burner and virtual numbers are provisioned through wholesale Voice over IP (VoIP) providers such as Peerless Network, Bandwidth, and Telnyx. Unlike traditional phone lines, these numbers can be obtained programmatically through an app, require no physical SIM card, and can be activated and discarded within minutes [23]. They are commonly classified as "nonFixedVoIP" in carrier lookup systems. Caribbean NANP numbers from jurisdictions such as Jamaica (876), the U.S. Virgin Islands (340), and Trinidad and Tobago (868) use the same dialing format as domestic U.S. numbers but may terminate at international or premium rates. This enables one-ring fraud, where attackers induce victims to return missed calls to revenue-generating numbers [22].

Identifying the carrier behind a phone number normally requires the Number Portability Administration Center (NPAC) [7], a restricted database that law enforcement accesses through subpoena [14]. Commercial APIs such as Twilio Lookup v2 draws from the same underlying sources and returns the carrier name and line

[1] https://800notes.com/

type for any queried number [27]. We note here that toll-free numbers are managed through the RespOrg system rather than through standard carrier databases, which means commercial APIs often returns no carrier identity for toll-free numbers.

2.2 Prior Studies

Prior telephony-based scam studies differ primarily in their measurement vantage point. That vantage point determines which part of the scam workflow is observable. Carrier-side work uses call detail records or signaling metadata and therefore has strong visibility into call volume, timing, and network relationships. Tu et al. [25] systematized telephony spam defenses and identified caller-ID spoofing as a central challenge. Li et al. [9] applied machine learning to carrier records from a major Chinese operator using temporal and graph-derived features. These approaches provide network-scale metadata, but they generally do not observe victim-described callback instructions, spoofed-number claims, or semantic relationships expressed in complaint narratives.

Honeypots trade population coverage for content. Gupta et al. [6] introduced Phoneypot as an early telephony honeypot. Prasad et al. [17] scaled this approach to tens of thousands of lines and used audio fingerprinting to identify robocall campaigns. SnorCall [18] extended this line of work with weak-supervision NLP over robocall transcripts, and later work combined academic and industry honeypots, enforcement data, and consumer complaints to cross-validate measurements over multiple years [19]. Honeypots provide audio and transcript content, but they require active infrastructure and usually cover shorter windows than long-running public complaint corpora.

Behavioral work measures how users respond to telephony threats and defenses. Tu et al. [26] showed that users remain susceptible to spoofed calls even under warning conditions. Sherman et al. [24] measured user responses to anti-robocall application indicators, and Pandit et al. [16] explored virtual assistants as a mediation layer for robocall handling. At the fraud-mechanism level, Sahin et al. [23] systematized telephony fraud, and later work quantified how premium-rate international routes can be monetized through International Revenue Share Fraud (IRSF) [22]. These studies explain user exposure and attacker incentives, but they do not recover complaint-derived relationships among phone numbers.

2.3 Complaint Data as Infrastructure Intelligence

A complaint page is titled with the reported phone number (e.g., "1-315-547-3793") and contains a list of user-submitted comments from victims who received calls from that number. Each comment includes free-text describing the victim's experience, the date of the report, and a screen name. Some comments contain only a few words ("spam call, do not answer"), while others describe the full scam interaction, including additional phone numbers the victim was told to call back, numbers that appeared on their caller ID, and numbers that texted them. These embedded cross-references are extracted by **EXPOSE**. The platform also auto-links mentioned phone numbers as hyperlinks to their respective complaint pages, but does not label why the number was mentioned. The context labels used in this paper (callback, spoofed, SMS, other) are defined in Table 1.

Prior work has treated complaint data as a reputation score for individual numbers. Pandit et al. [15] evaluated telephony blacklists built from complaint volumes and estimated their effect on unwanted calls, but treated each reported number independently. Miramirkhani et al. [12] studied technical-support scams by tracing web infrastructure that funneled victims into scam call centers, but did not analyze relationships between phone numbers inside victim complaints. The FTC Consumer Sentinel Network aggregates fraud reports for enforcement, but public analyses are aggregate summaries rather than complaint-level structural analysis [4].

In other security domains, researchers have shown that individual identifiers may look noisy or disposable in isolation, but the relationships between them can expose a more stable operational structure. Rahbarinia et al. [20] used DNS-query relationships to discover malicious domains connected to known infrastructure. Meiklejohn et al. [11] showed that transaction relationships can cluster pseudonymous Bitcoin addresses into real-world entities. Our work applies the same insight to telephony complaints. We treat victim narratives not as isolated reports about individual numbers, but as a source of typed cross-reference observations between numbers. The resulting framework uses these relationships to uncover the hidden callback and shadow infrastructure that victims describe in their complaints but never report as standalone complaint pages.

3 EXPOSE: FRAMEWORK DESIGN

Overview. We introduce **EXPOSE** (**Ex**posing **P**atterns **o**f **S**hadow Scam **E**cosystems), a four-stage pipeline that turns raw open-source complaint narratives into actionable infrastructure intelligence (Figure 1). The framework requires only three inputs (a reported phone number, free-text complaints, and timestamps) and is not tied to any single complaint platform. Stage 1 (Section 3.1) finds phone numbers in complaint text and labels each as callback, spoofed, SMS, or other. Stage 2 (Section 3.2) connects reported numbers to mentioned numbers in a directed graph, partitioning them into frontline, bridge, and shadow roles. Stage 3 (Section 3.3) queries carrier and line-type metadata for each callback number. Stage 4 (Section 3.4) groups numbers into campaigns by text similarity and links campaigns into shadow scam ecosystems through shared backend indicators. Each stage consumes the output of the previous one, and the full pipeline runs without manual intervention once the corpus is provided.

3.1 Stage 1: Extract and Label

Victim complaints often contain more than one phone number, but existing complaint platforms discard these embedded references entirely. They index only the reported number and treat each complaint page as an isolated record. **EXPOSE** treats them as structured evidence. scanner applies regular expressions to each comment to identify phone-number strings, normalizes matches to E.164 format, and pairs each with the complaint page number. Self-references (where the victim restated the page number) are removed because they carry no cross-reference value. **EXPOSE** labeler examines a 120-character window around each mentioned number and tests it

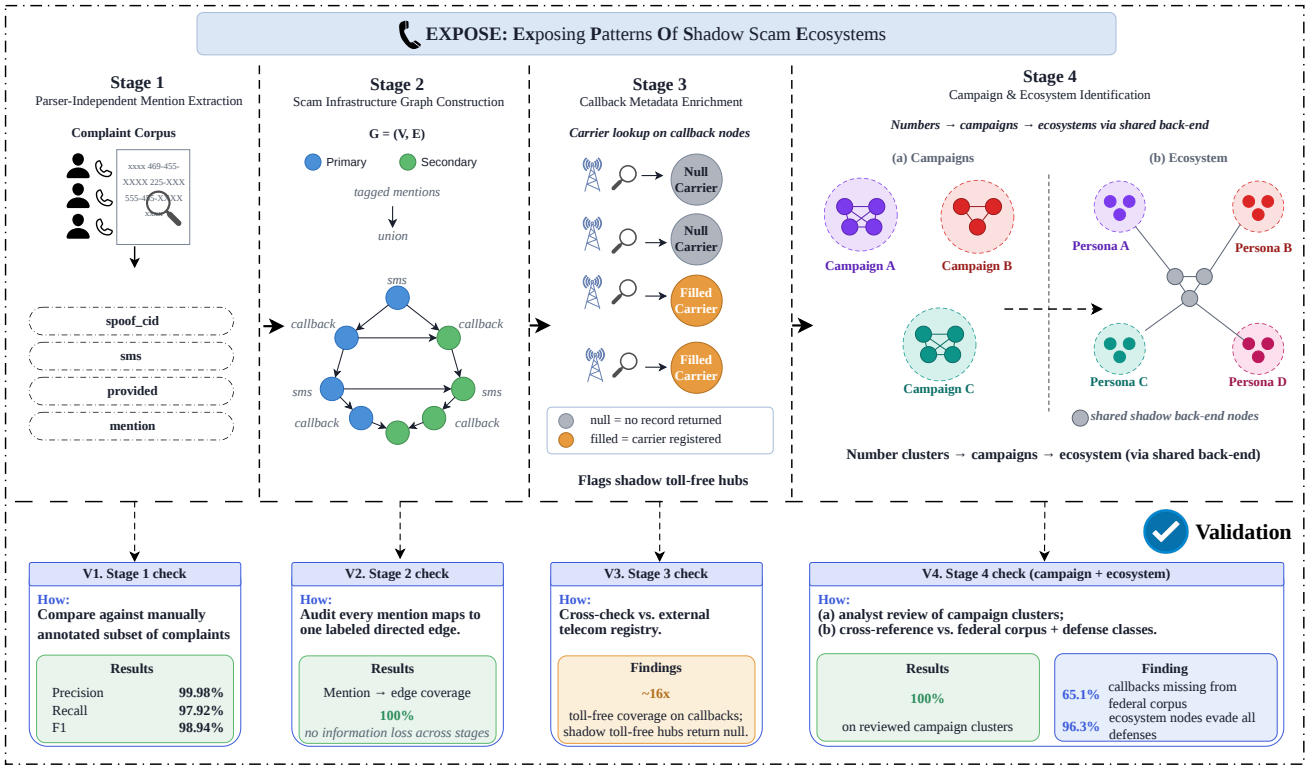

Figure 1: Analytical pipeline and data dependencies.

against 27 regular-expression patterns in priority order (full list in Appendix C, Table 6).

Label validation. Two authors independently label 200 stratified mentions. On the three functional labels (`callback`, `spoofed`, `sms`), agreement is 103/137 ($\kappa = 0.63$, substantial [8]). Disagreement concentrates on the `callback`/`other` boundary, where complaint language is genuinely ambiguous. We resolve `callback` disagreements through discussion. Against consensus, the labeler achieves callback precision = 0.87, recall = 0.70, and `spoofed`/`sms` both exceed $F_1 = 0.95$.

3.2 Stage 2: Cross-Reference Graph Construction

Stage 1 produces labeled observations of the form (u, v, κ, τ). Stage 2 collects them into a weighted directed graph where each observation becomes an edge from reported number u to mentioned number v, weighted by supporting comment count. Connected components are identified via breadth-first search [1] on the undirected projection.

The graph partitions numbers into three roles(formalized in Section 4): *frontline* (has complaint page, never mentioned elsewhere), *bridge* (has complaint page, also mentioned by others), and *shadow* (appears only in complaint text, no complaint page). A shadow number labeled `callback` is a shadow callback number, the paper's primary object of study.

3.3 Stage 3: Carrier Metadata Analysis

Stage 2 identifies which numbers are callback infrastructure. Stage 3 asks how they are provisioned. In the United States, carrier identification requires the restricted NPAC database [7], accessed by law enforcement through subpoena [14]. We use the Twilio Lookup v2 API [27], specifically its `/v2/PhoneNumbers/` endpoint with the `line_type_intelligence` package, which returns carrier name, line type, and burner/virtual flag. This is the closest public approximation to subpoena-backed carrier identification. All results are point-in-time (March 2026) and used for characterization only, not attribution. Section 5 reports the findings.

3.4 Stage 4: Discover Campaigns and Ecosystems

Stages 1-3 operate at the phone-number level. Stage 4 lifts this to campaigns and shadow scam ecosystems.

Scam operators reuse the same script across many numbers. The **EXPOSE** campaign detector concatenates each number's complaints, strips phone-number strings, converts the result to a TF-IDF vector [10] over unigrams and bigrams, and links numbers whose cosine similarity exceeds $\tau = 0.8$. Each connected group is one campaign. We use TF-IDF over Sentence-BERT [21] because campaign identity here is defined by exact lexical fingerprints (persona names, dollar amounts), not semantic similarity. Nahapetyan et al. [13], and Prasad et al. [18] reach the same design conclusion for SMS phishing and robocall transcripts, respectively.

Different campaigns may share backend infrastructure, meaning the same operator runs them. The **EXPOSE** ecosystem linker connects campaigns using three indicators of decreasing strength. A shared callback number (σ_{hub}) is the strongest when two campaigns directing victims to the same backend are unlikely to be unrelated. A complaint cross-reference (σ_{edge}) is second, where a number from one campaign appears in the complaints of another. A shared carrier (σ_{carrier}) is weakest; the same wholesale carrier is circumstantial because wholesale providers serve many unrelated tenants. We build a primary graph using only σ_{hub} and σ_{edge}, and an augmented graph adding σ_{carrier}. Our strongest claims come from the primary graph. Section 6 reports the results.

Framework Validation. We validate **EXPOSE** on a public 800notes corpus comprising 93,008 reported phone numbers and 97,093 victim comments posted between February 2007 and March 2026, spanning 19 years of complaint activity. 800notes is one of the longest-running open complaint platforms in the United States[2].

4 RQ1: THE CROSS-REFERENCE SIGNAL

Section 3.2 showed that victims often mention phone numbers beyond the one they are reporting. These extra numbers arrive in two ways: spoken during the call, when the caller asks the victim to dial back on a different number, or in a follow-up SMS. In both cases, the reported number is the contact surface and the extra number is the actual callback endpoint. We now formalize this indicator, instantiate it on the validation corpus, and evaluate the hidden callback layer it exposes.

4.1 Complaint Cross-Reference Graph Construction

EXPOSE begins by converting complaint text into typed phone-number cross-references. For each complaint attached to a directly reported phone number u, the extraction pipeline identifies zero or more complaint-mentioned phone numbers $v \neq u$ and assigns each mention a context label. Each non-self mention yields one complaint-derived observation

$$\omega = (u, v, \kappa, \tau),$$

where u is the directly reported number, v is the phone number mentioned in the complaint text, κ is the mention context, and τ is the comment timestamp. We remove self-references, where $v = u$, because they repeat the page identifier and do not add cross-reference information.

We write

$$\Omega = \{(u, v, \kappa, \tau)\}$$

for the resulting multiset of non-self complaint-derived observations. Ω preserves evidence at the comment level: if multiple comments mention the same source–target pair, those mentions remain separate observations until graph aggregation.

Let

$$R = \{u_i\}$$

denote the set of directly reported phone numbers, i.e., numbers with complaint pages in the corpus. Let

$$M = \{v : \exists (u, v, \kappa, \tau) \in \Omega\}$$

[2]https://800notes.com/

Table 1: Complaint-derived observations by context label ($|\Omega| = 9{,}756$; self-references removed).

Context κ	Count	%	Meaning
`callback`	842	8.6	Victim instructed to call this number
`spoofed`	309	3.2	Caller ID showed a different number
`sms`	282	2.9	Text came from this number
`provided`	339	3.5	Number given during the interaction
`mention`	7,984	81.8	General narrative reference

denote the set of phone numbers recovered from the complaint text. R captures the visible page-level view, while M captures the additional numbers victims mention inside narratives.

EXPOSE constructs a weighted directed cross-reference graph

$$G = (V, E, w),$$

where

$$V = R \cup M,$$
$$E = \{(u, v) : \exists (u, v, \kappa, \tau) \in \Omega\},$$

and

$$w(u, v) = |\{(u, v, \kappa, \tau) \in \Omega\}|.$$

The edge $u \to v$ records that complaints about u mention v. The edge direction represents the complaint-described relationship from the reported number to the mentioned number; it does not imply call routing, legal ownership, or common operator control. Context labels are retained at the observation level, while $w(u, v)$ counts the number of complaint comments supporting the aggregated source–target pair.

Instantiated on the validation corpus, the extraction pipeline observes 17,948 raw phone-number mentions. Of these, 8,192 are self-references and are removed. The remaining 9,756 complaint-derived observations form Ω and collapse to 8,542 unique directed source-target pairs. The edge-induced subgraph contains 9,316 graph-participating nodes. When isolated, directly reported numbers are included, the full graph contains 98,681 nodes.

Table 1 summarizes the context labels assigned to observations in Ω. These counts are complaint-level observations, not unique target counts; for example, the 842 callback observations collapse to 666 unique callback targets analyzed later.

This graph construction separates the visible complaint-page view from the numbers that appear only inside victim narratives. The next subsection uses R and M to define the node roles that make hidden callback infrastructure visible.

4.2 Role Partition and Illustrative Component

The sets R and M induce three observable roles in the **EXPOSE** graph:

$$V_F = R \setminus M, \qquad V_B = R \cap M, \qquad V_S = M \setminus R.$$

Here, V_F contains frontline numbers, which are directly reported numbers that are never mentioned elsewhere in complaint text. V_B contains bridge numbers, which have their own complaint pages and also appear as mentioned numbers in other complaints. V_S contains shadow numbers, which appear only inside complaint text and have no standalone complaint page in the corpus.

This role partition separates the visible complaint-page layer from the infrastructure recovered only through victim narratives.

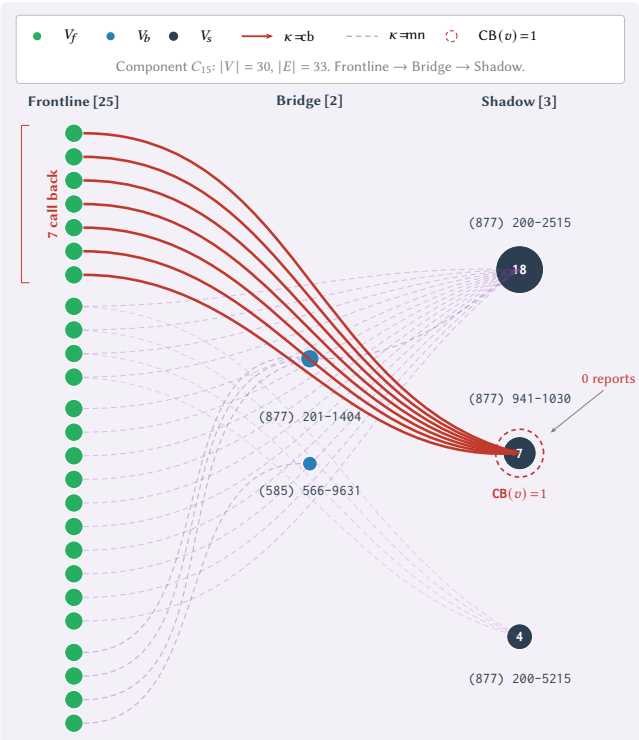

Figure 2: Illustrative **EXPOSE** component C_{15}. Open circles are frontline numbers with standalone complaint pages; gray diamonds are bridge numbers that are both directly reported and mentioned elsewhere; black squares are shadow numbers that appear only inside complaint text. Solid arrows denote `callback` observations and dashed gray edges denote lower-specificity mention observations. The dashed ring marks the shadow callback target in this component.

In the instantiated graph, **EXPOSE** identifies 92,065 frontline numbers, 943 bridge numbers, and 5,673 shadow numbers. Of the 9,756 complaint-derived observations in Ω, 7,916 (81.1%) point to shadow numbers. Thus, most complaint-described phone-number relationships lead to identifiers that are absent from page-level complaint views.

Figure 2 illustrates this role partition on component C_{15}, one of the larger connected components in the **EXPOSE**. The component contains 25 frontline nodes, 2 bridge nodes, and 3 shadow nodes. The figure is illustrative rather than evidentiary by itself; the aggregate claims in this section are based on corpus-level counts.

In the component, visible frontline pages converge on a small number of hidden targets. One shadow node is a callback target referenced by seven distinct frontline pages through `callback` observations, while the other shadow nodes are connected through weaker mention observations. This example shows why page-level complaint ranking misses part of the infrastructure: the visible frontlines appear as independent complaint pages, but the reused target exists only because victims wrote it inside complaint narratives.

The role partition is the first indication that complaint narratives expose infrastructure beyond the reported caller ID. The next subsection focuses on the callback-specific subset of this shadow layer, where the framework identifies numbers that victims were explicitly instructed to call.

4.3 Lurking Callback Targets

The role partition identifies which numbers are visible as complaint pages and which numbers exist only inside complaint text. We next focus on the callback subset of this structure, because callback mentions provide the clearest complaint-derived signal of backend telephony infrastructure. A callback target is a number that a victim says they were instructed to call.

Let

$$C_{\text{cb}} = \{v : \exists(u, v, \texttt{callback}, \tau) \in \Omega\}$$

denote the set of unique callback targets. These are unique phone numbers, not callback observations. Thus, the 842 callback observations in Table 1 collapse to a smaller set of unique callback targets.

Callback targets can appear in two roles. If a callback target also has its own complaint page, we call it a bridge callback target:

$$C_{\text{bridge}} = C_{\text{cb}} \cap V_B.$$

If a callback target appears only inside complaint text and has no standalone complaint page in the corpus, we call it a lurking callback target:

$$C_{\text{lurk}} = C_{\text{cb}} \cap V_S.$$

This distinction is central to **EXPOSE**. Bridge callback targets are visible to page-level complaint systems because they have complaint pages. Lurking callback targets are not visible in that way; they are recovered only because victims mention them inside complaints about other numbers.

In the corpus, **EXPOSE** uncovers

$$|C_{\text{cb}}| = 666$$

unique callback targets. Of these,

$$|C_{\text{lurk}}| = 541$$

are lurking callback targets and

$$|C_{\text{bridge}}| = 125$$

are bridge callback targets. Thus, most callback targets that victims were instructed to call have no standalone complaint page in the corpus.

For each callback target v, we measure reuse by the count of distinct reported numbers that reference it:

$$d(v) = |\{u : \exists(u, v, \texttt{callback}, \tau) \in \Omega\}|.$$

We use $d(v)$ rather than raw comment count because repeated comments attached to the same complaint page should not inflate the apparent reuse of a backend target. A high value of $d(v)$ means that multiple visible complaint pages direct victims to the same callback number.

For example, suppose complaints about three different reported numbers u_1, u_2, u_3 each state that victims were told to call the same number v:

$$(u_1, v, \texttt{callback}, \tau_1), \quad (u_2, v, \texttt{callback}, \tau_2), \quad (u_3, v, \texttt{callback}, \tau_3).$$

Then $d(v) = 3$. If $v \notin R$, the number has no standalone complaint page and therefore belongs to C_{lurk}. A page-level complaint system

Table 2: Top-10 lurking callback targets by count of distinct reported numbers. Targets are masked for presentation.

Rank	Masked target	Count	Source area codes	Type
1	877-***-0760	15	9	Toll-free
2	877-***-3699	10	1	Toll-free
3	877-***-1030	7	6	Toll-free
4	307-***-9758	6	4	Geographic
5	833-***-2573	6	5	Toll-free
6	866-***-0072	5	2	Toll-free
7	855-***-0740	4	2	Toll-free
8	304-***-1294	3	2	Geographic
9	833-***-2576	3	3	Toll-free
10	833-***-2577	3	3	Toll-free

cannot rank v by direct complaint volume, but **EXPOSE** uncovers it by aggregating callback references inside complaint narratives.

The callback layer also differs from the visible complaint-page layer in its numbering structure. In the directly reported corpus, 3.3% of numbers are toll-free. Among callback targets, 363 of 666 (54.5%) are toll-free, a 16.5× enrichment over the directly reported baseline. The corresponding contingency table yields $\chi^2 = 4{,}895$ with $df = 1$ and $p \ll 0.001$. We interpret this result conservatively: the observed asymmetry is consistent with functional specialization, where visible frontlines and callback targets occupy different roles in the scam workflow.

Table 2 reports the ten highest-ranked lurking callback targets by $d(v)$. Targets are masked for presentation. The top lurking callback target is referenced by 15 distinct reported numbers spanning 9 source area codes, yet it has no standalone complaint page in the corpus.

This subsection establishes the main hidden-infrastructure result. Many callback numbers are not directly reported, yet multiple visible complaint pages point to them. The next subsection evaluates whether the resulting graph structure is coherent and whether these lurking callback targets can be surfaced in an operational setting.

4.4 Validation and Operational Value

The previous subsection showed that **EXPOSE** uncovers lurking callback numbers that victims were instructed to call, but that have no standalone complaint page. We now evaluate whether lurking callback hubs can be detected without hindsight, and whether **EXPOSE** adds visibility beyond complaint-volume baselines.

Can shadow callback hubs be detected as they emerge? We tested **EXPOSE** can flag shadow callback hubs as they appear, without access to future data. We test this by processing callback observations in chronological order. For each shadow callback number v, the detector maintains the running count $d(v)$ of distinct frontline pages that have referenced it. When $d(v)$ reaches a threshold, the detector fires an alert. The logic is simple. If three unrelated victims, complaining about three different frontline numbers, all independently write "call back at" the same number, that convergence is unlikely to be coincidental. The detector is a counter-based rule applied to each callback observation as it arrives.

We use $d(v) \geq 3$ because it is the lowest threshold at which all flagged hubs are confirmed in manual review. We split the corpus at January 1, 2024, treating 2007 through 2023 as history and 2024 through 2026 as the evaluation window. All 15 shadow callback hubs that reach $d(v) \geq 3$ in the full chronology also cross the threshold during the evaluation window. Median detection latency is 6 days. Six of 15 are detected the same day, and 8 of 15 within one week.

Table 3: Lurking callback target recovery: **EXPOSE** versus complaint-volume blacklist.

Method	Targets	Lurking targets found	Leverage
Blacklist ≥ 1 complaint	38,046	0 / 541	1.0×
Blacklist ≥ 5 complaints	3,091	0 / 541	1.0×
Blacklist ≥ 10 complaints	1,395	0 / 541	1.0×
EXPOSE top-10 lurking hubs	10	10 / 541	5.6×

Figure 3 traces one hub (877)-941-1030 step by step. Two victims on two different frontline pages both write "call back at (877)-941-1030" on the same day, bringing $d(v)$ to 2. Forty-seven days later, a third victim on a third page writes the same instruction, pushing $d(v)$ to 3 and firing the alert. At that point, the hub still has zero direct complaints and is invisible to the frontline complaint page. Four subsequent observations confirm continued activity. The bottom panel of Figure 3 shows the latency distribution across all 15 hubs.

Comparison with complaint-volume baselines. Finally, we compare **EXPOSE** against a page-level baseline: ranking numbers by the number of direct complaints on their own complaint page. This baseline reflects how complaint-driven reputation systems commonly operate. However, it cannot recover a lurking callback target by construction, because a lurking target has no standalone complaint page in the corpus.

EXPOSE uncovers 541 lurking callback targets. A complaint-volume blacklist recovers 0 of these 541 targets at every complaint threshold. This is not a thresholding artifact; the baseline lacks the information needed to surface nodes that appear only inside complaint narratives. We define leverage as the number of distinct visible frontlines surfaced per ranked target. The ten highest-ranked lurking callback hubs account for 56 distinct visible frontlines, whereas a top-10 complaint-volume blacklist affects only ten directly reported frontlines by construction.

Taken together, these evaluations show that the **EXPOSE** is not merely a visualization of complaint text. It uncovers coherent scam components, supports precision-oriented early warning for lurking callback hubs, and exposes backend callback targets that page-level complaint-volume methods cannot observe.

5 RQ2: CARRIER ENRICHMENT AND ANALYSIS

Section 4 established which numbers serve as shadow scam backend infrastructure by recovering 666 callback targets (541 lurking, 125 bridge). This section addresses how is the shadow callback layer provisioned, and whether existing carrier-level defenses can identify shadow scam infrastructure? We answer the question here and feed the result into Section 6 as the σ_{carrier} operation-linking indicator.

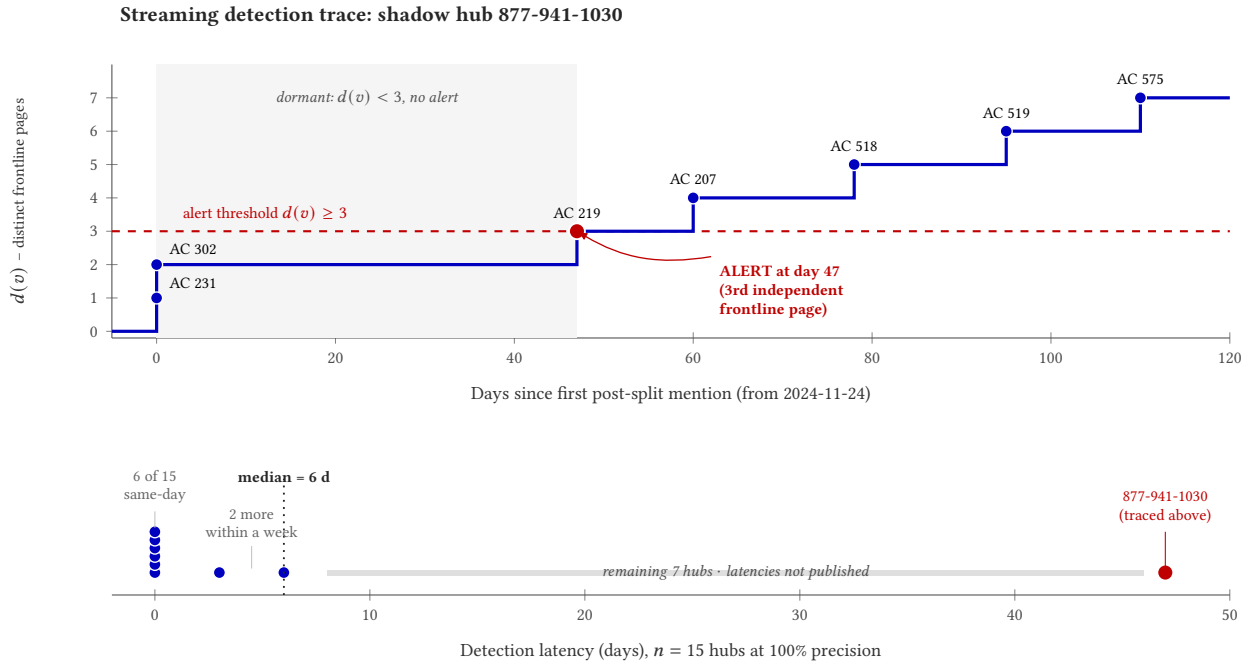

Figure 3: Top: streaming-detection trace for shadow callback hub 877-941-1030. Independent victims complaining about distinct frontline numbers each contribute one observation toward $d(v)$; the alert fires on the third independent page (day 47), at which point the hub still has zero direct complaints. Bottom: latency distribution across the 15 hubs that reached $d(v) \geq 3$ in the post-split evaluation window. Six hubs were detected the same day; eight cumulatively within one week; median latency 6 days. The traced hub is an outlier at the upper end of the published range.

We query the Twilio Lookup v2 API [27] for each callback target. The API draws from the same underlying number-portability databases (Number Portability Administration Center (NPAC), LERG) that law enforcement accesses via subpoena [2, 14], and returns a line-type classification, carrier name when resolvable, and a burner/virtual flag. All 666 queries returned successfully. Carrier assignments reflect provisioning state at lookup time (March 2026); numbers may have been ported since their involvement in scam activity. We report all findings as observed under this specific API and do not claim they generalize to all carrier identification systems.

5.1 Carrier Enrichment Results

We report three findings in sequence: what type of numbers scam operators use, which carriers provision them, and where carrier identification breaks down.

Finding 1: Toll-free and VoIP dominate. Measurement shows that toll-free numbers account for 54.5% of all callback targets (363/666), a 16.5× enrichment over the 3.3% toll-free share of the directly reported corpus ($\chi^2 = 4{,}895$, $p \ll 0.001$). NonFixedVoIP, i.e., burner or virtual accounts for another 22.4% (149/666). Together, these two categories cover 76.8% of callback infrastructure. Mobile (8.3%) and landline (9.0%) are minorities.

Finding 2: No major consumer carrier provisions shadow infrastructure. For the 249 shadow geographic callback targets, all top-10 carriers returned by the API are wholesale or programmable VoIP providers (Figure 4): Peerless Network (8.0%), Onvoy (7.6%), Bandwidth (5.6%), and others. T-Mobile, Verizon, and AT&T are absent.

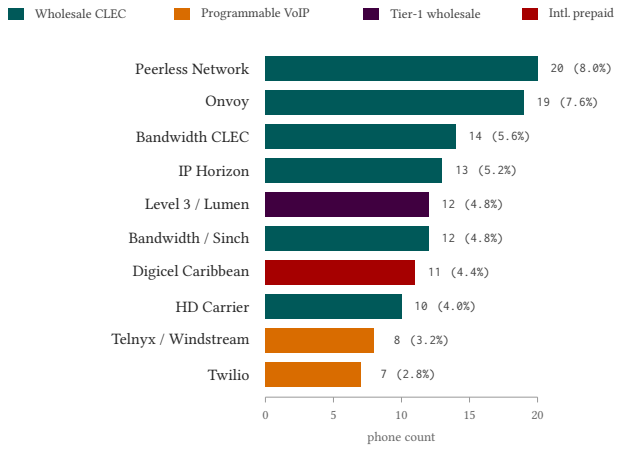

Figure 4: Carrier identity for the 249 shadow geographic callback numbers (Twilio Lookup v2). Bars are colored by carrier type: wholesale CLECs and aggregators dominate (Peerless, Onvoy, Bandwidth, IP Horizon, HD Carrier); the remainder are programmable VoIP platforms (Telnyx, Twilio), one tier-1 wholesale provider (Level 3 / Lumen), and one international prepaid carrier (Digicel Caribbean,the Jamaica-876 cluster). No major consumer carrier appears in the top-10 nor in the long tail. Among these 249 numbers, 47.4% (118) also carry the burner / virtual flag, indicating programmatic provisioning with no physical SIM.

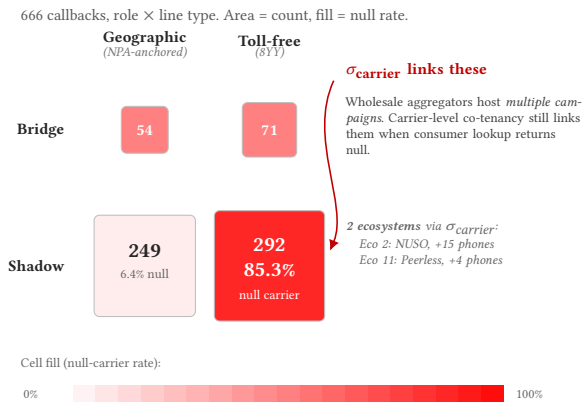

Figure 5: The carrier-opacity cliff. Cross-tabulation of 666 callback numbers by structural role × line type. Cell area is proportional to count; cell fill encodes null-carrier rate from Twilio Lookup v2.σ_{carrier} uncovers two ecosystems (NUSO, Peerless).

Among these 249 numbers, 118 (47.4%) carry the burner/virtual flag, indicating programmatic provisioning with no physical SIM. A distinct cluster of 12 callback targets carries Jamaican area code 876, all provisioned by Digicel Caribbean on prepaid SIM cards with fan-in of 1, consistent with the Jamaica lottery scam topology documented by Tu et al. [25].

Finding 3: Toll-free carrier identification breaks down. Among 292 shadow toll-free callback targets, 249 (85.3%) return a null carrier name. For bridge targets, the null rate is 50.4% (63/125). For shadow geographic targets, it drops to 6.4% (16/249). This gradient reflects how U.S. toll-free numbers are provisioned: they are managed through the Somos Responsible Organization (RespOrg) system rather than through the carrier databases that serve geographic numbers. The RespOrg identity is not exposed through standard telecom lookup APIs. Among the 43 shadow toll-free targets where a carrier is resolved, Bandwidth/Zipwhip (24, 55.8%) and Twilio (9, 20.9%) account for 76.7%, both programmable platforms with instant number provisioning. We note that this null-carrier result is specific to the Twilio API and may reflect provisioning opacity at the API level. Carrier-internal databases or direct RespOrg queries through the SMS/800 system may resolve some of these numbers. We report it as an operational visibility limitation under one widely used commercial API. Figure 5 cross-tabulates this gradient by structural role and line type: shadow toll-free phones are simultaneously the largest cohort (292) and the most opaque (85.3% null), forming the "cliff" that consumer-facing lookup cannot see across. We return to how (σ_{carrier}) recovers ecosystems on this cliff in Section 6.3.

5.2 Implications for Operation Linking

The carrier analysis produces two outputs that feed directly into the operation-linking stage (Section 6). First, the carrier name itself becomes a linking indicator. If two campaigns that share no complaint-derived edges (no shared callback hub, no cross-reference) both route their callback traffic through the same wholesale carrier, this provisioning overlap is circumstantial evidence of shared supply-chain infrastructure. We formalize this as $\sigma_{\text{carrier}}(\mathcal{K}_i, \mathcal{K}_j) = 1$ when callback targets from two campaigns share the same dominant carrier. This is the weakest of the three operation-linking indicator. In Section 6.5, we show that carrier-only links produce the only false positive and partial match among the 12 recovered operations, confirming that carrier similarity should be treated as secondary evidence. Second, the carrier enrichment results explain why certain defenses fail at the operation level. The absence of major consumer carriers (Finding 2) means that carrier-side abuse teams at T-Mobile, Verizon, and AT&T have no visibility into the shadow callback layer. The null-carrier rate for toll-free numbers (Finding 3) means that toll-free traceback requires RespOrg-level investigation rather than standard carrier lookup. Both of these structural gaps are quantified in the defense-gap analysis of Section 6.7.

6 RQ3: EXPOSING SHADOW SCAM ECOSYSTEMS

This section presents the results of **EXPOSE**'s fourth stage and answers our third research question. grouping phone numbers into campaigns and linking campaigns into shadow scam ecosystems. Sections 4 and 5 operate at the level of individual phone numbers. But scam operations are not organized at that level. Frontline numbers are rotated frequently, while scripts and callback backends are reused across many numbers. To capture that higher-level structure, we introduce a three-level hierarchy.

6.1 Three-Level Hierarchy

We adapt the message→campaign→operation model of Nahapetyan et al. [13] to the telephony complaint setting with two differences. First, the observable unit here is a human-authored complaint narrative rather than a machine-generated SMS template. Complaint text is therefore noisier and more variable, which makes exact string matching insufficient and motivates the use of a statistical similarity model [10]. Second, the infrastructure indicators in our setting are telephony-native, such as shared callback hubs, shared carrier-level provisioning signatures, and direct cross-references, rather than web-native indicators such as URLs or hosting artifacts.

Definition 6.1 (Three-Level hierarchy). Level 1 (Identifier): each phone number is an atomic unit. *Level 2 (Campaign):* a campaign $\mathcal{K}_i$ is a connected component in a thresholded complaint-similarity graph over phone numbers. Numbers in the same component share a recurring behavioral pattern, such as a script, persona, or claimed affiliation. *Level 3 (Ecosystem):* a shadow scam ecosystem O_j is a group of campaigns that share backend telephony infrastructure despite using different scripts.

Level 2 captures what victims hear or read. Level 3 captures what runs behind it. Two numbers belong to the same campaign because their complaints sound alike. Two campaigns belong to the same ecosystem because they direct victims to the same callback backend. Without this hierarchy, text analysis separates scripts but misses shared backends, and infrastructure analysis sees backends but cannot distinguish scripts. Both are needed.

6.2 Level 2: Identifying Persona-Based Campaigns

We group phone numbers into campaigns by comparing their complaint text. For each phone number with at least 50 characters of complaint text ($n = 3{,}537$ numbers), we concatenate all associated comments into one document $x(u)$, truncated to 1,000 characters to prevent prolific reporters from dominating the representation. We strip phone-number strings before comparison so that grouping reflects what was said, not which numbers were mentioned. We represent each document as a TF-IDF vector [10] over unigrams and bigrams:

$$\mathbf{d}_u = \mathrm{tfidf}(x(u)).$$

We use TF-IDF [10] for three reasons. First, complaint narratives are short, sparse, and formulaic. Scam scripts reuse the same persona names, dollar amounts, and phrasing across dozens of phone numbers. In this setting, lexical overlap is sufficient to identify repeated scripts, as demonstrated by Nahapetyan et al. [13], who cluster SMS phishing messages into 35,128 campaigns using near-identical content matching on similarly short and templated text. Second, TF-IDF is deterministic and interpretable, which allows threshold sensitivity analysis ($\tau = 0.7, 0.8, 0.9$) and manual review of campaign boundaries. Third, it requires no campaign-level supervision, which does not exist for telephony complaints. We select TF-IDF over embedding-based alternatives such as Sentence-BERT [21] because embeddings capture semantic similarity (e.g., "loan" and "credit" are close), whereas campaign identity in our domain is defined by exact lexical fingerprints (e.g., the persona "Sarah" with "$82,000" is a different campaign from "Morgan" with "$85,000"). Prasad et al. [18] reach a similar conclusion for robocall transcripts, using weak-supervision NLP with lexical features rather than learned embeddings.

We then construct a thresholded similarity graph

$$G_\tau = (U, E_\tau),$$

where

$$(u_i, u_j) \in E_\tau \iff \cos(\mathbf{d}_{u_i}, \mathbf{d}_{u_j}) \geq \tau.$$

A campaign is defined as a connected component of G_τ. Let

$$\mathcal{K} = \{\mathcal{K}_1, \mathcal{K}_2, \ldots, \mathcal{K}_m\}$$

denote the set of campaigns recovered from G_τ, where each

$$\mathcal{K}_i \subseteq U$$

is a set of phone numbers. We report as campaigns only non-singleton connected components, since isolated phone numbers do not provide evidence of repeated script reuse.

At the operating threshold $\tau = 0.8$, this procedure uncovers 203 non-singleton campaigns containing 608 phone numbers. The largest campaign contains 26 phone numbers; the median campaign contains 2. The remaining phone numbers with sufficient text are isolated in G_τ and are therefore not counted as campaigns.

Threshold τ controls the trade-off between over-merging and fragmentation. At $\tau = 0.7$, the method yields 280 non-singleton campaigns; at $\tau = 0.9$, it yields 152. We use $\tau = 0.8$ as the operating point because it balances lexical strictness and campaign coverage while preserving the higher-level infrastructure links used in the next stage.

This campaign layer captures what victims report hearing or reading. For example, complaints using recurring personas such as "Sarah," "Morgan," and "Olivia" remain distinct campaigns when their wording, named personas, and monetary claims differ systematically, even when they all fall under the broader category of loan-related fraud. The next stage asks a different question: when do distinct campaigns nonetheless belong to the same backend operation, even though they reuse telephony infrastructure?

6.3 Linking Campaigns into Shadow Scam Ecosystems

Campaigns capture what victims hear. The next question is whether distinct campaigns share backend infrastructure, meaning the same operator runs them. We link campaigns into ecosystems using three indicators. The strongest is a shared callback number (σ_{hub}), where two campaigns direct victims to call the same backend number, making it unlikely they are unrelated. The second is a complaint cross-reference (σ_{edge}), where a phone number from one campaign appears in the complaint text of another. The weakest is a shared carrier (σ_{carrier}), where callback numbers from two campaigns are provisioned through the same wholesale carrier, which is circumstantial because wholesale providers serve many unrelated tenants.

To separate complaint-derived evidence from carrier evidence, we build two ecosystem graphs. The primary graph uses complaint-derived indicators:

$$(\mathcal{K}_i, \mathcal{K}_j) \in E_\Sigma^{\text{primary}} \iff \sigma_{\text{hub}} = 1 \ \vee \ \sigma_{\text{edge}} = 1. \tag{1}$$

The augmented graph adds carrier links:

$$(\mathcal{K}_i, \mathcal{K}_j) \in E_\Sigma^{\text{aug}} \iff \sigma_{\text{hub}} = 1 \ \vee \ \sigma_{\text{edge}} = 1 \ \vee \ \sigma_{\text{carrier}} = 1. \tag{2}$$

An ecosystem is a connected component in the corresponding graph. The primary graph comprises 10 ecosystems, spanning 24 campaigns and 72 phone numbers (81 total nodes, including 9 shadow callback hubs). The augmented graph comprises 12 ecosystems, spanning 32 campaigns and 101 phone numbers (111 total nodes, including 10 shadow hubs). Figure 6 shows all 12 shadow scam ecosystems uncovered by **EXPOSE**.

6.4 Indicator Coverage

Table 4 evaluates whether each indicator contributes unique information. No single indicator uncovers all operations. σ_{edge} alone finds 7; σ_{hub} contributes 3 that are invisible to both other indicators; σ_{carrier} adds 2 more. The hub-linked operations are the most distinctive output of **EXPOSE**. They rely on shared callback infrastructure that is not visible in page-level complaint counts or in carrier metadata alone. The carrier indicator generates the most edges (10) but discovers only 3 ecosystems, because many carrier edges connect campaigns already linked by σ_{edge}.

6.5 Evaluation

We evaluate every recovered operation by reading the full complaint text of each constituent campaign and judging whether the campaigns plausibly reflect the same organizational actor. We followed one specific criterion. Campaigns must share specific operational indicators (same callback number, same persona, consistent dollar amounts, aligned temporal behavior, or the same reporter tracking

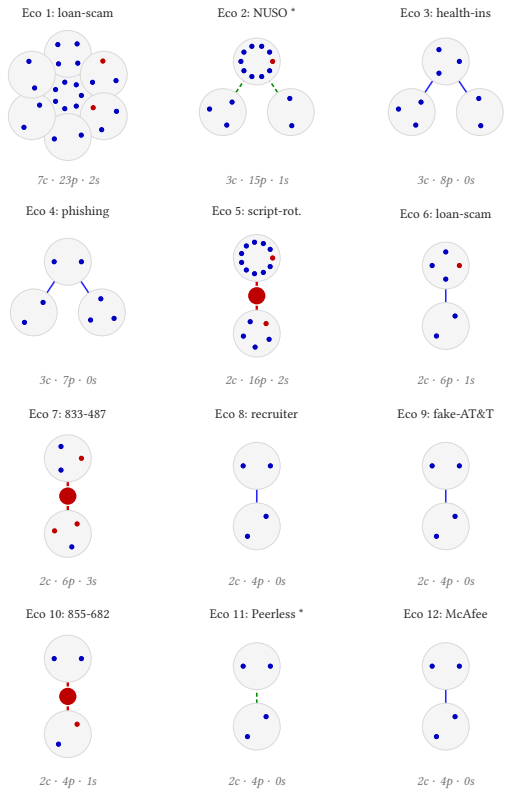

Figure 6: All 12 shadow scam ecosystems uncovered by **EXPOSE** (32 campaigns, 101 phones, 10 shadow callback hubs). Node colors: • frontline, • bridge, • shadow. Edge styles: red solid = σ_{hub}, blue solid = σ_{edge}, green dashed = $\sigma_{carrier}$. Asterisked panels are augmented-only (carrier-linked). Footer reports campaigns · phones · shadows.

one evolving scam), not merely a broad scam category such as "loan scam."

Primary graph (complaint-derived evidence only). Using only σ_{hub} and σ_{edge}, the pipeline recovers 10 operations, all of which are correct under exhaustive manual review i.e.; **100% precision** (10/10). All three hub-discovered operations are correct; all seven edge-discovered operations are correct. Complaint-derived callback reuse and cross-references are high-precision signals for lifting campaigns into operations.

Table 4: Each row shows an indicator configuration, how many campaign-level links and ecosystems it produces, and how many ecosystems are correct under exhaustive manual review.

Signal configuration	Links	Eco.	Correct	Precision
σ_{hub} only	3	3	3	100%
σ_{edge} only	11	7	7	100%
σ_{carrier} only	10	3	1	33.3%
Primary ($\sigma_{\text{hub}} + \sigma_{\text{edge}}$)	14	10	10	100%
Augmented (all three)	24	12	10	83.3%

Augmented graph (adding carrier similarity). When σ_{carrier} is added, the pipeline expands to 12 operations. Ten are correct, one is a false positive (two unrelated campaigns linked only because both callback hubs are provisioned through the wholesale carrier Peerless Network), and one is a partial match (a Google-business campaign grouped with two IRS campaigns via NUSO co-tenancy). Overall precision: **83.3%** (10/12). The false positive and partial match are both caused exclusively by σ_{carrier}, confirming that carrier similarity should be treated as secondary evidence (Table 4).

6.6 Case Study: Script Evolution on Shared Infrastructure

Ecosystem 5 in Appendix D Table 8 illustrates why the hierarchy is necessary. Campaign TC_0004 (11 phones) uses the persona "Jennifer," presents as an underwriter, and directs victims to callback hub 877-200-3699. Campaign TC_0013 (5 phones) initially uses the persona "Marcos," presents as a funding department, and offers up to $60,000. At the script layer, these campaigns remain distinct (cosine similarity 0.12). The ecosystem graph reconnects them through a second callback hub, 877-396-0066, which appears in both campaigns. One TC_0004 phone references it directly. One TC_0013 phone page, originally reported for the Marcos script, later receives a Jennifer-script complaint from the same reporter that also references 877-396-0066. The shared hub has zero direct complaints and is visible only through callback cross-references.

This pattern reveals not just infrastructure convergence but script rotation on shared infrastructure: the same operator runs the Jennifer and Marcos scripts through overlapping callback backends, reusing phone pages across persona transitions. A text-only analysis correctly separates the two scripts but cannot detect that they share backend infrastructure. The **EXPOSE** uncovers the link.

6.7 Defense Gap Analysis

We model four defense classes. **D1** (complaint-volume blacklist) catches a frontline number if it has at least five direct complaints. **D2** (caller-ID authentication relevance) catches a frontline number if **EXPOSE** observes it in at least one spoofed-caller-ID cross-reference; this measures complaint-visible spoofing, not whether STIR/SHAKEN would block the call. In callback scams, the victim initiates the return call, so caller-ID authentication on the originating leg does not address callback infrastructure. **D3** (major-carrier abuse visibility) catches a shadow hub if its Twilio Lookup carrier maps to a major retail carrier with a public abuse-response pathway. **D4** (toll-free lookup visibility) catches a toll-free shadow hub if Twilio returns a non-null carrier. The D4 result is specific to Twilio Lookup v2 and may reflect API-level provisioning opacity; corroborating it with a second lookup source would strengthen the finding. We therefore interpret it as an operational visibility limitation under one widely used commercial API.

Table 5 shows that 78 of 81 nodes in the primary graph (96.3%) and 100 of 111 nodes in the augmented graph (90.1%) fall outside every modeled defense. Nine of ten primary ecosystems have a 100% defense gap. Under our visibility model, D3 and D4 catch zero shadow hubs because the relevant hubs are provisioned through wholesale VoIP carriers or the toll-free RespOrg system rather than through directly visible retail-carrier paths.

Table 5: Defense gap across both operation graphs.

	Augmented (111 nodes)		Primary (81 nodes)	
Defense	Catches	Cov.	Catches	Cov.
D1: Blacklist (≥5)	3/101	3.0%	3/72	4.2%
D2: Spoofed CID	8/101	7.9%	0/72	0%
D3: Major carrier	0/10	0%	0/9	0%
D4: TF traceback	0/10	0%	0/9	0%
Any defense	11/111	9.9%	3/81	3.7%
Gap	**100/111**	**90.1%**	**78/81**	**96.3%**

Finding 4: The defense gap is architectural, not operational. The 96.3% gap does not mean that existing defenses fail at their intended tasks. Each targets a different layer: blacklists catch high-complaint frontlines, caller-ID authentication addresses the originating call, major-carrier abuse workflows monitor visible subscribers, and toll-free traceback depends on carrier visibility. EXPOSE shows that the recovered backend layer sits outside these views: operators rotate frontlines, victims initiate callbacks, and shadow hubs are concentrated behind wholesale VoIP or toll-free provisioning paths. The gap is therefore a structural mismatch between defense design and scam architecture.

Finding 5: Targeting the hidden backend has higher leverage than targeting the visible frontline. A complaint-volume blacklist blocks one directly reported number at a time. EXPOSE instead ranks shadow callback hubs that serve as backends for multiple frontlines. In the primary ecosystems, targeting the top shadow hubs yields 10.1× leverage compared with 1.0× for complaint-volume ranking: the same number of interventions, directed at backend callback hubs rather than visible frontlines, affects an order of magnitude more observed infrastructure.

6.8 FTC Cross-Validation

The 541 lurking callback numbers are one of the core findings, but is this invisibility specific to our dataset, or does it hold in other complaint systems? We test this by cross-referencing all 666 shadow callback numbers and 541 lurking callback numbers against the FTC public Do Not Call complaint files covering five fiscal years (FY 2021-2025) and totaling 8,540,146 records. Of the 666 shadow callback numbers, 394 (59.2%) have zero matches in the FTC corpus, and of the 541 lurking callback numbers, 352 (65.1%) are similarly absent. These numbers are invisible not just on 800notes but also across 8.54 million federal complaints. For comparison, the 125 bridge callback numbers (which do have complaint pages on 800notes) are FTC-invisible at a lower rate of 33.6%, consistent with the structural prediction that numbers with no victim reported directly on one platform are also less likely to appear in a different complaint system. Of the 541 shadow numbers, 290 were first seen within the FTC reporting window (October 2020 through September 2025). This does not prove shadow infrastructure is absent from all regulatory systems, but it confirms that the invisibility of shadow callback numbers persists in a large, independent federal complaint source and is not an artifact of our dataset.

7 LIMITATIONS

This study has five limitations. First, empirical validation uses one public corpus, 800notes, so the results support recoverability and internal coherence rather than cross-corpus universality. We partially address platform specificity by cross-referencing recovered callback targets against public FTC Do Not Call complaints, but this is an external visibility check, not a full second-corpus instantiation of **EXPOSE**. Second, complaint data is voluntary and self-selected, so we do not estimate population-level prevalence. Third, graph edges represent complaint-described relationships, not verified call routing, legal ownership, or common control. Fourth, Twilio line-type and carrier metadata are point-in-time observations and should be interpreted as lookup-time characterization rather than historical provisioning. Fifth, precision estimates are tied to the reviewed samples or components in each evaluation, and recall for hidden infrastructure cannot be estimated without independent ground truth.

EXPOSE is portable by input contract. It requires a reported identifier, free-text narratives, and timestamps. In principle, the same abstraction could be instantiated on other abuse-reporting domains where victims describe secondary identifiers, such as email addresses, URLs, wallet addresses, or social-media handles. This paper validates the approach only on telephony complaints; cross-corpus and cross-domain validation remain future work.

8 CONCLUSION

Existing defenses against telephony fraud operate on individual phone numbers, yet scam operations are multi-number organizations that hide persistent backend callback hubs behind disposable frontline numbers. A defense that sees only the frontline misses the backend entirely. However, victim complaint narratives often describe these hidden numbers, including callback targets, spoofed caller-IDs, and SMS sources, but existing complaint platforms discard them as unstructured text. Our framework **EXPOSE** recovers this overlooked signal and uses it to uncover hidden scam infrastructure from open-source complaints alone.

We validated **EXPOSE** on a public 800notes corpus of 93,008 reported phone numbers and 97,093 comments spanning 19 years. **EXPOSE** identified 541 lurking callback numbers that victims were explicitly instructed to dial, yet that have no standalone complaint page anywhere in the corpus. The shadow layer is structurally distinct from the visible frontline. Toll-free numbers account for 54.5% of callback targets versus 3.3% of reported numbers, a 16.5x enrichment, and 85.3% of shadow toll-free hubs return no carrier identity under Twilio Lookup v2. Notably, no major consumer carrier provisions any numbers in this layer; provisioning concentrates entirely in wholesale VoIP and the toll-free RespOrg system. Lifting numbers into higher-level structures exposes 203 persona-based campaigns and 12 shadow scam ecosystems, the majority of which fall outside every defense class we model. Cross-validation against 8.54 million FTC complaints confirms that 65.1% of shadow callback targets are invisible to the federal complaint system as well. To support ongoing efforts in exposing and combating scams, we release the **EXPOSE** framework and its associated dataset publicly.

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A OPEN POLICY

In support of open science and independent verification, we have released the full EXPOSE-Toolkit and all derived artifacts required to reproduce every quantitative claim in this paper at the following anonymous repository:
https://anonymous.4open.science/r/EXPOSE-Toolkit-87E3/

The 800notes validation corpus (93,008 reported numbers, 97,093 comments, February 2007 – March 2026) is released as results.jsonl. All records are drawn from publicly posted complaint pages; no private, authenticated, or scraped-behind authentication content is included. Reporter screen names are removed; only publicly posted phone numbers and narrative text are retained. An accompanying FTC Do-Not-Call comparison uses FTC bulk-release public records; we include a fetch script rather than redistributing those files. External enrichment. Carrier metadata (Section 5) is obtained from the Twilio Lookup v2 API. We release a cached lookup file covering all 666 callback targets as of March 2026 so reviewers can reproduce Section 5 with a Twilio account; a refresh script is provided for users with API credentials.

Reproducibility. A single entry point (scripts/reproduce_all.sh)

B ETHICAL CONSIDERATIONS

This study uses publicly accessible complaint data from 800notes.com and public FTC Do Not Call complaint files. We did not place calls, send messages, interact with suspected operators, or attempt to trigger scam infrastructure. We also did not collect, infer, or analyze the real-world identities of complainants. Our objective is to recover complaint-derived infrastructure relationships, not to attribute activity to specific individuals or organizations.

Although the underlying complaint pages and phone numbers are publicly visible, they may still contain sensitive or misleading information. Complaint text can include user error, subjective interpretation, and references to numbers that are spoofed or otherwise not under the scammer's control. For this reason, we treat the findings as a representation of complaint-described relationships rather than as proof of legal ownership or common operational control. We avoid publishing raw complaint text beyond short excerpts necessary for explanation, and we mask phone numbers in tables and figures unless the exact value is essential to the argument.

Our telecom enrichment is also passive. Twilio Lookup v2 is used only to retrieve line-type and carrier metadata for callback targets already present in the complaint corpus. We use these results as

infrastructure characterization signals, not as attribution evidence. Likewise, FTC matching is used only to test whether recovered callback targets are visible in an independent public complaint source; it is not used to infer enforcement status or regulatory awareness.

C LABEL PATTERN DICTIONARY

Table 6 lists all 27 keyword patterns used by the rule-based context classifier in Stage 1 (Section 3.1). A mention is assigned the label of the first matching pattern found in a 120-character window around each mentioned number. If no pattern matches, the label defaults to `other`.

Table 6: All 27 keyword patterns used for context labeling, grouped by label. Each pattern is matched against a 120-character window around each mentioned number.

#	Label	Pattern
1	`callback`	call back at
2	`callback`	call back to
3	`callback`	call us at
4	`callback`	call us back
5	`callback`	call them at
6	`callback`	call me at
7	`callback`	please call
8	`callback`	reach us at
9	`callback`	reach them at
10	`callback`	give us a call
11	`callback`	dial
12	`callback`	return the call to
13	`spoofed`	caller id showed
14	`spoofed`	caller id said
15	`spoofed`	caller id was
16	`spoofed`	displayed as
17	`spoofed`	came up as
18	`spoofed`	showed up as
19	`spoofed`	number showed
20	`spoofed`	appeared as
21	`sms`	text from
22	`sms`	texted me
23	`sms`	text message from
24	`sms`	got a text
25	`sms`	got a message from
26	`sms`	sms from
27	`sms`	sent a text
–	`other`	(default: no pattern matched)

D CAMPAIGN AND ECOSYSTEM EVIDENCE

This appendix provides supporting evidence for the campaign identification and ecosystem discovery results presented in Section 6. Table 7 reproduces representative complaint scripts for the 10 largest campaigns. Table 8 provides verbatim excerpts and manual evaluation verdicts for all 12 discovered ecosystems. Table 9 lists persona and dollar-amount fingerprints across loan-scam campaigns. All complaint text is reproduced verbatim from publicly posted 800notes.com pages.

Table 7: Top-10 campaigns by phone count. Each campaign is identified by TF-IDF cosine similarity ≥ 0.8 across complaint narratives. Persona, dollar amount, and callback number (where present) serve as the behavioral fingerprint. Callback numbers shown in full; frontline numbers redacted.

ID	Persona	Phones	Area Code	Representative complaint text (verbatim, truncated)
TC_0000	Matt Bieneman†	26	3	"Unknown robocall scammer. Calls once or twice each weekday using a different forged Caller ID number each time. Apparently always unassigned phone numbers. Has been going on for several months."
TC_0001	"Sarah"	17	7	"Hi. This is Sarah with an important update from our approval team. According to my records, it looks like you may be approved for a consolidation loan of up to $82,000. Payments for these loans can start as low as $480 per month. Please give me a call as soon as possible at `1-877-200-1765`."
TC_0002	"Morgan"	16	12	"Hello. This is Morgan with an important update from our loan eligibility group. According to my system, you may be qualified for a consolidation loan of up to $85,000. Payments for these loans can start as low as $480 per month. Please give me a call as soon as possible at `1-877-200-2515`."
TC_0003	"Olivia"	14	8	"Hi. This is Olivia. I'm a senior loan processor with our approval department. According to our system, you may pre-qualify for a consolidation loan of up to $47,000. Payments for this loan start as low as $350 a month. Please give us a call back at `877-200-0760`."
TC_0004	"Jennifer"	11	1	"Hi, this is Jennifer. I'm a senior underwriter on the debt resolution team. Great news. I have a note here that you are prequalified for a consolidation loan of up to $75,000. Payments can be as low as $430 a month. These offers are time sensitive, so please give us a call back at `1-(877)-200-3699`."
TC_0005	"Violet"	10	7	"Hey this is Violet Rodriguez reaching out from the account prescreen team. I see that we are able to get you preapproved for a personal loan starting at $85,000 with a manageable monthly payment of $700 per month. Please note that approval slots are filling quickly. Let's connect at `833-487-2563`."
TC_0006	(Google scam)	10	1	"Another cover number used by google business scam robo giving a call back of `619-935-1051` to opt out . . . caller ID showed 'Des Moines Ia' "
TC_0007	(Comenity Bank)	9	5	"This is the 64th number used by Comenity Bank. They have called 233 times since 3/28/19 and have never left a message."
TC_0008	(Loan scam)	7	2	"Callback number for a very persistent automated loan scam system that calls almost every day outside of Sundays (leaving a voicemail if you miss or reject the call), always from a different spoofed number, so blocking calls & reporting them does no good. . . "
TC_0009	"Jennifer"	6	6	"Hi, this is Jennifer. I'm an underwriter with our approval department. I'm just calling to let you know it appears you've been pre-qualified for a personal loan of up to $72,000 with payments starting as low as $530 a month. Please give us a call back here `877-941-1030`."

TC_0000 is a single reporter documenting 26 spoofed caller-ID numbers from the same robocall operation.
TC_0009 uses the same persona name "Jennifer" as TC_0004 but a different dollar amount ($72K vs. $75K), different title ("underwriter" vs. "senior underwriter"), and a different callback number (`877-941-1030` vs. `877-200-3699`). TF-IDF correctly separates them (cosine = 0.71, below $\tau = 0.8$).

Table 8: All 12 ecosystems with linking evidence. For each ecosystem, we show the signal that produced the link, the shared entity (callback hub shown in full; carrier shown by name), and verbatim excerpts from two constituent campaigns. "Eval" is the manual correctness judgment from Section 6.

Eco	Signal	Shared entity	Campaign A excerpt	Campaign B excerpt
1	carrier+edge	HD Carrier LLC	*TC_0008* (7ph): "Callback number for a very persistent automated loan scam system that calls almost every day outside of Sundays, always from a different spoofed number…"	*TC_0021* (4ph): "Loan scam again, this making it the 27th call since August 18th (only skipping Sundays, Labor Day & September 26th). The call was rejected when I saw the area code…"
2	carrier	NUSO	*TC_0006* (10ph): "Another cover number for google business scam robo giving a call back of 619-935-1051 to opt out"	*TC_0045* (3ph): "Didn't answer and message left for an IRS audit scam requesting call back to 844-988-2564."
3	edge	(cross-ref)	*TC_0040* (3ph): "Health Insurance Coverage, 800-491-5500, 270-012-8867, 833-536-0586 yourcoverage.net"	*TC_0054* (3ph): "Ally Commercial, small Business Funding, FAX ad"
4	edge	(cross-ref)	*TC_0176* (2ph): "Fraudulent ROBOCALL that is for a SCAM loan application that is FAKE and this company does not exist. This is a PHISHING SCAM…"	*TC_0200* (2ph): "Got another unsolicited pending loan approval call, which I did not apply for. Of course, all they would need is for me to give them all of my sensitive information."
5	**hub**	`+1-877-396-0066`	*TC_0004* (11ph): "Hi, this is Jennifer. I'm a senior underwriter on the debt resolution team…prequalified for a consolidation loan of up to $75,000…call back at 1-(877)-200-3699."	*TC_0013* (5ph): "Hello. This is Marcos from the funding department. It looks like you may qualify for up to $60,000…Please call 1-877-201-1356."
6	edge	(cross-ref)	*TC_0026* (4ph): "Call received at 6:04 PM on 10/9/2025, the call was rejected when I saw the area code (expecting it to be another loan scam attempt)."	*TC_0181* (2ph): "This is a scam loan call. The calls come from different numbers daily (red flag #1). The calls started right after a data breach at Stellantis Financial…"
7	**hub**	`+1-833-487-2573`	*TC_0052* (3ph): "Loan processing department robo scam. Call back number 833-487-2573"	*TC_0053* (3ph): "Loan processing department robo scam. Call back numbers 833-487-2573 or 833-487-2576 or 833-487-2577…"
8	edge	(cross-ref)	*TC_0111* (2ph): "Sent unwanted group text claiming to be an Indeed recruiter"	*TC_0078* (2ph): "Call-back number referenced in an unwanted text from (229) 722-8136, claiming to be an Amazon recruiter"
9	edge	(cross-ref)	*TC_0113* (2ph): "They are not really AT&T (they are a scam call center in India [no calls during the Diwali holiday]). Verified this with the real AT&T…"	*TC_0141* (2ph): "These people call every weekday. They do not leave a message. When I call back from another number they claim they are AT&T Wireless…"
10	**hub**	`+1-855-682-0740`	*TC_0125* (2ph): "Hi, this is Lauren calling from the approval department regarding your prequalified personal loan for $91,000…call back at 855-682-0740."	*TC_0126* (2ph): "This is Lauren calling from the approval department regarding your prequalified personal loan for $91,000…call us back at 855-682-0740."
11	carrier	Peerless Network	*TC_0137* (2ph): "Received this message that started out as what sounded like an Indian lady…A company name was not provided…"	*TC_0173* (2ph): "Fraudulent robocall claiming you have a tax debt and must call 888-516-0672 back to resolve the issue."
12	edge	(cross-ref)	*TC_0159* (2ph): "Scam email containing a fake invoice for a McAfee 48-month Antivirus Pro subscription for $510.22. Provides this number for 'support'…"	*TC_0174* (2ph): "Scam email containing a fake invoice for a 4-year McAfee 'Ultimate ShieldX' subscription for $338.00. Provides this number for 'Customer Service'…"

Table 9: Persona-dollar fingerprints in loan-scam campaigns. TF-IDF separates campaigns with different fingerprints despite shared scam category. Two "Jennifer" variants (TC_0004 and TC_0009) use different amounts and callback numbers.

Campaign	Persona	Amount	Callback hub
TC_0001	Sarah	$82,000	877-200-1765
TC_0002	Morgan	$85,000	877-200-2515
TC_0003	Olivia	$47,000	877-200-0760
TC_0004	Jennifer	$75,000	877-200-3699
TC_0005	Violet	$85,000	833-487-2563
TC_0009	Jennifer	$72,000	877-941-1030
TC_0013	Marcos	$60,000	877-201-1356
TC_0125	Lauren	$91,000	855-682-0740
TC_0126	Lauren	$91,000	855-682-0740